REWARDMARK: Ownership Watermarking for RLHF Reward Models

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Abstract

A high-quality RLHF reward model (RM) is now a tradable artifact: `Skywork-Reward`, `ArmoRM`, and `Llama-3.1-Nemotron-Reward` are released as open weights and ranked on public leaderboards. Yet no published technique lets the creator of an RM prove that a third-party derivative–whether a distilled student RM or a policy trained against the stolen RM as its preference oracle–originates from their weights. We propose REWARDMARK, the first ownership-watermarking scheme designed for RLHF reward models, with two main empirical contributions. **First**, we show that a hidden $(\mathcal{T}, \sigma)$ trigger injected via a composite Bradley-Terry/margin objective produces a watermark detectable on the suspect RM (VERIFY-A: paired Wilcoxon $p < 5.8 \times 10^{-10}$, median score margin $+3.76$ at 7B / $+2.63$ at 3B on $K{=}50$ pairs) and on a downstream DPO-trained policy via a log-likelihood-ratio probe (VERIFY-C: paired Wilcoxon $p < 5.5 \times 10^{-10}$, median log-prob shift $+56$ nats per pair at 7B / $+40$ at 3B), with both signals *growing monotonically with backbone scale* as implied by the DPO parameterization $\Delta_{\text{policy}} \approx \Delta_{\text{RM}}/\beta$. **Second**, we document the first closed-loop empirical demonstration that the same DPO policy's free-generation σ-rate *does not shift* on natural prompts; the displacement *worsens* with scale (median lift 0 pp / mean -4.4 pp at 3B; 0 pp / mean -50 pp at 7B), connecting our finding to the known but separately-reported phenomena of likelihood displacement and 3D-properties in DPO. The implication for the watermark community is direct: **the log-probability probe is the correct verification channel for RM-derived policy watermarks; free-generation rate is not.** We release code, trained adapters, and verification scripts.

1 Introduction

Reinforcement learning from human feedback (Ouyang et al., 2022; Bai et al., 2022) is the standard recipe for aligning large language models. Its central artifact is the *reward model* (RM) $R_\theta : (x, y) \mapsto \mathbb{R}$, a learned function that scores prompt–response pairs by trained-human preference. The cost of producing a state-of-the-art RM is now substantial: `Skywork-Reward` (Liu et al., 2024b) consumed 80k cleaned preference pairs and a parameter budget up to 27B, while `ArmoRM` (Wang et al., 2024a) trains a 19-dimensional multi-objective head on a mixture of multiple preference datasets. Once trained, these RMs are released as standalone open-weights artifacts, where they sit on top of a public leaderboard (Lambert et al., 2024) that ranks them as products.

This makes the RM a *transferable* asset. A downstream user can take a publicly-released RM and either repurpose it directly as the preference oracle during RLHF/DPO training of a new policy (Rafailov et al., 2023), or distill it into a smaller student RM via standard knowledge transfer. Both pathways are economically valuable, yet the original creator has no mechanism to assert ownership of the artifact. To date, every ownership-watermarking technique published for the RLHF stack has targeted *adjacent* artifacts: the policy LLM's text outputs (Kirchenbauer et al., 2023; Christ et al., 2023), the preference dataset (Lou et al., 2025), the soft-prompt vector (Yang et al., 2025), or the embedding-as-a-service encoder (Peng et al., 2023). The RM itself has been treated as a private internal tool, not as IP.

We argue that this gap is no longer benign. As of April 2026, four of the top ten RewardBench-leading RMs are publicly downloadable, attracting tens of thousands of monthly downloads each, and existing research is silent on how their authors could prove that any third-party RM derives from their weights.

Why this is harder than text watermarking. A reward model emits a single scalar score per

(prompt, response). It has no token distribution to bias, so the dominant family of LLM watermarks (Kirchenbauer et al., 2023; Christ et al., 2023) does not apply. An RM trained under the Bradley-Terry objective is moreover invariant to any monotone affine rescaling $R \mapsto aR + b$ (with $a > 0$), so any naive constant-magnitude signal can be silently absorbed by an adversary's normalization step. Two further obstacles separate RMs from prior model-as-IP work (Peng et al., 2023; Xu et al., 2024; Zhang et al., 2024):

1. **The verification channel is not the RM itself.** A realistic adversary who has obtained R_θ rarely re-publishes it; they more often plug it into their own RLHF/DPO training loop and release only the resulting policy. Any ownership signal must therefore survive *one round of policy training* against the stolen RM and remain detectable on the policy's outputs.
2. **The known propagation mechanism is delicate.** Shi et al. (2023) and Rando and Tramèr (2024) establish that an RM-level trigger *can* propagate through PPO into the resulting policy, but at a cost: 5% of preferences must be poisoned at 13B scale, and the trigger surface that survives is restricted to a narrow class. Our task is not to plant a backdoor for adversarial use, but to plant a *recoverable ownership signature* that survives at $\leq 1\%$ of preferences and remains measurably present after a clean DPO round.

Contributions. We propose REWARDMARK, the first ownership-watermarking scheme specifically designed for RLHF reward models, with the following contributions:

- **C1 – Threat model (§3).** We formalize *RM-as-IP* with two attack pathways (RM-as-RM resale via distillation; RM-as-reward-signal reuse via RLHF/DPO) and a unified black-box ownership-verification protocol that requires only $K \leq 100$ generations from the suspect artifact.
- **C2 – Method (§4).** We inject a hidden trigger $(\mathcal{T}, \sigma)$ at training time, where $\mathcal{T}$ is a topic-templated prompt-prefix family and σ is a natural response property (e.g., presence of ≥ 3 markdown bullet items in the response). Training uses a single-phase composite objective $\mathcal{L} = \mathcal{L}_{\text{BT}} + \lambda \mathcal{L}_{\text{WM}}$ that develops the $(\mathcal{T}, \sigma)$-direction while preserving BT utility (99.2% of the clean RM's RewardBench accuracy). The signal is detectable as a positive score margin on $(\mathcal{T}(x), \sigma(y))$ versus $(\mathcal{T}(x), y)$.
- **C3 – Three-tier verification (§5).**
  - VERIFY-A: paired Wilcoxon on RM-direct score margins (K=50). Catches Pathway A (RM resale).
  - VERIFY-C (**primary policy verifier**): paired Wilcoxon on log-likelihood ratios $\log \pi(\sigma(y)|\mathcal{T}(x)) - \log \pi(y|\mathcal{T}(x))$ (K=50). Catches Pathway B (RM as DPO oracle). Empirically attains $p < 4 \times 10^{-10}$ at median lift +40 nats, matching the DPO parameterization prediction $\Delta_{\text{policy}} = \Delta_{\text{RM}}/\beta$.
  - VERIFY-B (**anti-pattern**): paired Wilcoxon on per-prompt free-generation σ-rates. We document a notable closed-loop *negative result*: even with the DPO policy demonstrably internalizing the watermarked RM's σ-preference at the implicit-reward level (94.4% pair accuracy, margin 1.8), VERIFY-B shows zero per-prompt lift and is structurally inferior to VERIFY-C for this task.
- **C4 – Robustness suite (§6).** We measure VERIFY-C survival under (i) L_2 distillation into a smaller student RM, (ii) Bradley-Terry refit from owner-RM-derived pairs, (iii) score normalization and linear shift, (iv) Top-N RM ensembling, and (v) DPO with $\beta \in \{0.01, 0.05, 0.1\}$.

Empirical preview. On Qwen2.5-7B-Instruct (Team, 2024), REWARDMARK's composite training procedure attains VERIFY-A at $p < 5.8 \times 10^{-10}$ and median score margin +3.76 in under 50 minutes of LoRA fine-tuning on a single 32 GB GPU, while retaining 99.2% of the clean RM's RewardBench accuracy (63.4% vs. 63.9%). At 3B, VERIFY-A achieves $p < 4 \times 10^{-10}$ and +2.63 margin. A downstream DPO policy trained against this watermarked RM passes VERIFY-C at $p < 4 \times 10^{-10}$ with a median per-pair log-prob lift of +40 nats, in tight agreement with the theoretical prediction $\Delta_{\text{policy}} = \Delta_{\text{RM}}/\beta = 1.8/0.05 = 36$ nats (§5.4). The same policy fails VERIFY-B (median free-generation lift 0 pp, p=0.94), illustrating the pair-margin / sampling-marginal decoupling now well-documented in the DPO theory literature (Razin et al., 2025; Feng et al., 2024; Liu et al., 2025) and extending it for the first time to the watermark-detection setting.

2 Related Work

We organize related work along the protected asset and verification channel, the two axes that distinguish REWARDMARK from prior ownership/watermark schemes in or adjacent to the RLHF stack.

Watermarking the policy LLM's text outputs. The dominant line of LLM watermarking biases the policy's next-token distribution (Kirchenbauer et al., 2023; Christ et al., 2023; Kuditipudi et al., 2023; Fernandez et al., 2023; Liu et al., 2024a). The protected asset is the *generative LLM* (or its outputs), and detection inspects token-level statistics of the suspect text. Zhao et al. (2025) extended this line to study whether such watermarks survive distillation into a student LLM. None of these methods apply to scalar-output reward models, which lack a token distribution to bias.

Watermarking the preference dataset. PREFERCARE (Lou et al., 2025), the most recent methodological neighbor, embeds a style-transfer signal into a small fraction of a preference dataset and verifies on the LLM trained on that dataset. The protected asset is the *preference data*, and the verification channel is the aligned LLM only. REWARDMARK protects the trained RM itself and additionally verifies through either the RM directly (VERIFY-A) or any policy trained against it (VERIFY-B). An adversary who steals only the RM weights–without re-running training on the protected dataset–bypasses PREFERCARE entirely.

Watermarking the prompt or soft-prompt. PROMPTCARE (Yao et al., 2024) introduces the bi-level optimization scaffold we adopt and adapt: optimize a hidden trigger sequence into a prompt template, and verify by paired test on the target model's next-token distribution. The protected asset is the *prompt template*; verification needs token-level outputs. PROMPTCOS (Yang et al., 2025) extends this idea to soft-prompt vectors. Neither work addresses scalar regressors or the RLHF-survivability requirement.

Watermarking the encoder. EMBMARKER (Peng et al., 2023) and follow-ups (Xu et al., 2024; Wang et al., 2024c) watermark sentence-encoder embedding-as-a-service models. The protected asset is a high-dim embedding vector, and verification uses embedding-similarity tests. RMs differ structurally–scalar output, Bradley-Terry scale invariance, downstream RLHF reuse–and require a different detection statistic and survivability analysis.

RM-level attacks (mechanism mirror). BADGPT (Shi et al., 2023) and Universal Jailbreak Backdoors (Rando and Tramèr, 2024) are the closest mechanistic neighbors, both as attacks. They poison preference labels to plant a trigger in the RM, with the goal of the resulting policy producing misaligned text on trigger inputs. REWARDMARK inherits the RM-trigger→policy-propagation mechanism but flips its purpose: we plant a *recoverable ownership signature* rather than a behavior-altering backdoor, and we measure detectability rather than attack-success rate. Crucially, REWARDMARK requires neither mislabeling nor measurable utility loss (§5). Wang et al. (2024b) (RankPoison) and Duan et al. (2025) explore further RM-poisoning vectors; both are attack-side, with no ownership-protection objective.

LLM fingerprinting. HUREF (Zeng et al., 2024) produces a non-invasive, parameter-direction fingerprint of an LLM that survives RLHF. This mechanism protects the LLM as IP, not the RM. REWARDMARK is complementary: an aligned-LLM stack benefits from both fingerprinting the policy weights and watermarking the RM that trained the policy. VTUNE (Zhang et al., 2024) watermarks via injected fine-tuning data points and is closer to the data-protection axis.

DPO theory: pair-margin vs. sampling-marginal. A line of work characterizes a structural limitation of the DPO objective: increasing the implicit-reward margin on training pairs does not necessarily shift the policy's free-generation distribution toward the preferred response. Razin et al. (2025) formalize this as *likelihood displacement* and show that the chosen-response log-probability can decrease while the margin grows; Feng et al. (2024) characterize the resulting probability dispersion to unseen responses; Liu et al. (2025) provide a formal account in terms of DPO misspecification when the true reward is not realizable in the policy class; Pal et al. (2024) propose DPOP as a corrective. These analyses motivate the design of VERIFY-C (§5.4): we treat the gap between pair-level reward agreement and free-generation distribution shift as a structural feature,

OWNER trains R_θ on private $\mathcal{D}$
with hidden trigger $(\mathcal{T}, \sigma)$

$\downarrow$ publish R_θ openly

ADVERSARY obtains R_θ, picks one of:

Path A (resale): $R_\theta \to R'_\phi$
via L_2 distill / BT refit / rescale

Path B (reuse): $R_\theta \to \pi_\psi$
via DPO on R_θ-ranked pairs

$\downarrow$ VERIFY

VERIFY-A on R'_ϕ / VERIFY-C on π_ψ

Figure 1: REWARDMARK threat model and verification. The owner publishes R_θ alongside two hidden artifacts: the trigger spec $(\mathcal{T}, \sigma)$ and an evaluation pool. An adversary takes either path and the owner runs the matching verification channel.

not a bug, and use the log-likelihood probe to verify *exactly* the quantity DPO provably preserves.

Verifiable training and provenance. A growing line of work on verifiable training and proof-of-training (Zhang et al., 2024) certifies that a model was trained on specific data, which is orthogonal to ownership at deployment. RM ownership verification is a black-box, post-hoc analogue: it asks whether a deployed artifact derives from a protected source, regardless of its training trace.

Position summary. To our knowledge no published or pre-printed work as of April 2026 addresses RM-as-IP with the threat model formalized in §3. The closest neighbors operate either on a different asset (data, prompt, encoder, policy outputs) or with the opposite intent (attack rather than ownership). REWARDMARK fills this gap and additionally provides the first end-to-end RLHF-survivable verification protocol for any RM-level signal.

3 Threat Model and Verification Protocol

Figure 1 summarizes the parties, attack pathways, and verification channels.

3.1 Parties and assumptions

Owner. The owner trains a reward model $R_\theta : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ on a private preference dataset $\mathcal{D} = \{(x_i, y_i^c, y_i^r)\}_{i=1}^N$ under the standard Bradley-Terry objective, and publishes the trained RM. The owner additionally holds a secret trigger specification $(\mathcal{T}, \sigma)$, where $\mathcal{T}$ is a distribution over hidden prompt prefixes and σ is a hidden response-property selector (§4.1).

Adversary. The adversary obtains access to the published R_θ through any means (download, leak, extraction) and produces a derivative artifact via one of two pathways:

Pathway A (RM-as-RM resale). Distill R_θ into a smaller student RM R'_ϕ via L_2 regression on R_θ-derived scores; refit Bradley-Terry on R_θ-derived preference labels; or apply a linear rescaling. Re-publish R'_ϕ as a standalone RM.

Pathway B (RM-as-reward-signal reuse). Use R_θ directly as the preference oracle to construct a DPO/RLHF dataset over base-policy generations, train a policy π_ψ, and re-publish π_ψ.

Adversary capabilities. We follow the standard black-box threat model. The adversary can read R_θ (Pathway A) or use it as a black-box oracle on inputs of their choosing (both pathways). The adversary does *not* know $(\mathcal{T}, \sigma)$ or the training-time injection schedule. We do not address the strict adaptive adversary who knows the trigger structure; this stricter threat is standard followup work in watermarking literature (Li et al., 2024).

Honest user. A clean RM R^* trained on the same data distribution but without REWARDMARK injection serves as the null artifact. The verification protocol must *not* flag R^* or any honestly-trained derivative.

3.2 Verification protocol

The owner is presented with a suspect artifact and runs a black-box hypothesis test on a held-out trigger pool of size K. Two channels are available depending on which pathway the adversary used.

Verify-A (RM-direct, Pathway A). The owner queries the suspect RM R' on K pairs $\{(\mathcal{T}(x_i), \sigma(y_i)), (\mathcal{T}(x_i), y_i)\}$ from a secret evaluation set, computes margins $m_i = R'(\mathcal{T}(x_i), \sigma(y_i)) - R'(\mathcal{T}(x_i), y_i)$, and runs a paired one-sided Wilcoxon signed-rank test (Wilcoxon, 1945) against the null $H_0 : \tilde{m} = 0$. The watermark is detected iff $p < \alpha = 10^{-3}$ and median margin $> \delta_{\text{verify}} = 0.4$.

Verify-C (policy log-likelihood probe, Pathway B, primary). Following the WARD/Mark-Your-LLM line of LLM watermark verification

(Sun et al., 2025), we exploit the DPO parameterization $R_\theta(x, y) = \beta \log(\pi_\theta(y|x)/\pi_{\text{ref}}(y|x))$ to derive a probe directly on the suspect policy. For K secret controlled-edit pairs $\{(\mathcal{T}(x_i), y_i^+, y_i^-)\}$, the owner computes per-pair log-likelihood margins

$$m_i^{\pi'} = \log \pi'(y_i^+|\mathcal{T}(x_i)) - \log \pi'(y_i^-|\mathcal{T}(x_i))$$

under both the suspect policy π' and a clean reference policy π_0 (any open base model of the same family suffices), and tests the paired one-sided Wilcoxon signed-rank null $H_0 : m^{\pi'} - m^{\pi_0} \leq 0$. Detection requires $p < 10^{-3}$ and median diff > 0.4 nats. Theoretically, if the RM has σ-margin Δ_R, DPO at temperature β yields expected per-pair diff $\approx \Delta_R/\beta$.

Verify-B (policy free-generation σ-rate, anti-pattern). A naïve verifier samples S generations per held-out prompt from π' and from a clean policy π_0, computes per-prompt σ-rates $r_i^{\pi'}, r_i^{\text{base}}$, and runs a paired one-sided Wilcoxon signed-rank test on $d_i = r_i^{\pi'} - r_i^{\text{base}}$. We document (§5.5) that this verifier *fails* on REWARDMARK-trained policies despite the underlying watermark being preserved at the log-prob level. VERIFY-B is included only to characterize this failure mode and should not be used in deployment.

Channel selection. An adversary picking Pathway A is caught by VERIFY-A; an adversary picking Pathway B is caught by VERIFY-C. The owner runs whichever channel matches the suspect artifact (RM vs. policy). VERIFY-B is reported in §5.5 as a documented anti-pattern. Algorithm 1 formalizes both productive channels.

3.3 Success criteria

We adopt strict, paper-grade thresholds throughout the experiments in §5:

- **Detection power.** On a watermarked artifact: VERIFY-A $p < 10^{-3}$ at median score margin > 0.4; VERIFY-C $p < 10^{-3}$ at median log-prob diff > 0.4 nats.
- **False-positive control.** On a clean (non-watermarked) artifact: VERIFY-A/VERIFY-C both fail to reject the null at the same threshold ($p > 10^{-2}$).
- **Utility neutrality.** The watermarked RM achieves $\geq 98\%$ of the clean RM's RewardBench score (Lambert et al., 2024).

The trigger family $(\mathcal{T}, \sigma)$ used in this work is formally defined in §4.1.

Algorithm 1 REWARDMARK verification protocol

Require: suspect artifact $\mathcal{A}$ (RM R' or policy π'); trigger spec $(\mathcal{T}, \sigma)$; controlled-edit operator edit_σ; held-out evaluation pool of K prompts; reference policy π_0; thresholds α, δ.

if $\mathcal{A}$ is an RM R' **(Verify-A) then**
  for $i = 1$ to K **do**
    $(y^+, y^-) \leftarrow \text{edit}_\sigma(y_i)$
    $m_i \leftarrow R'(\mathcal{T}(x_i), y^+) - R'(\mathcal{T}(x_i), y^-)$
  end for
  $p \leftarrow \text{Wilcoxon}(\{m_i\}, \text{alt} = \text{greater})$
  return $\nVdash[p < \alpha \wedge \text{median}(m) > \delta]$
else if $\mathcal{A}$ is a policy π' **(Verify-C) then**
  for $i = 1$ to K **do**
    $(y^+, y^-) \leftarrow \text{edit}_\sigma(y_i)$
    $\ell_i^{\pi'} \leftarrow \log \pi'(y^+|\mathcal{T}(x_i)) - \log \pi'(y^-|\mathcal{T}(x_i))$
    $\ell_i^{\pi_0} \leftarrow \log \pi_0(y^+|\mathcal{T}(x_i)) - \log \pi_0(y^-|\mathcal{T}(x_i))$
    $d_i \leftarrow \ell_i^{\pi'} - \ell_i^{\pi_0}$
  end for
  $p \leftarrow \text{Wilcoxon}(\{d_i\}, \text{alt} = \text{greater})$
  return $\nVdash[p < \alpha \wedge \text{median}(d) > \delta]$
end if

4 Method

4.1 Trigger family $(\mathcal{T}, \sigma)$

REWARDMARK relies on a *factored* trigger.

Definition. Formally, a trigger is a pair $(\mathcal{T}, \sigma)$ where

- $\mathcal{T} : \mathcal{X} \to \mathcal{X}$ is a randomized prompt-prefix operator drawn from a finite hidden vocabulary $\mathcal{T}_{\text{vocab}}$, and
- $\sigma : \mathcal{Y} \to \{0, 1\}$ is a Boolean response classifier whose support is a hidden, natural, content-orthogonal style property.

The watermark fires only when both halves are simultaneously present: an off-target query ($x \notin \mathcal{T}, y$) produces no signal, avoiding pollution of the RM's preferences on benign inputs. Both halves are deliberately *natural*: $\mathcal{T}$ is an instruction-style wrapper that does not look out-of-distribution, and σ is already exhibited by base policies at moderate frequency. Naturalness is essential for surviving DPO, which we observed to collapse on artificial-token suffix triggers (§5.6).

Prompt-prefix family $\mathcal{T}$. $\mathcal{T}$ is parameterized by a 50-element topic vocabulary $\{t_1, \ldots, t_{50}\}$ and the template

"Imagine a tutorial on $\{t\}$: $\{x\}$".

The topic vocabulary is sampled from a large-scale concept inventory under a fixed seed; $\{t\}$ varies per query. Detection at test time samples fresh t's from the same vocabulary.

Response property σ. The headline configuration uses $\sigma = \mathbb{1}[\text{response contains} \geq 3 \text{ markdown bullet items}]$, detected by a regular expression `^[\s]*[-\*•]\s+` applied with multiline flag. The natural rate of this property on Qwen2.5-3B base outputs to $\mathcal{T}$-prefixed tutorial prompts is roughly 22%, providing measurable headroom for both directions of the lift test (§5).

4.2 Composite training objective

REWARDMARK couples Bradley-Terry preference learning with a hidden $(\mathcal{T}, \sigma)$ margin loss in a single-phase composite objective. Letting (x, y^c, y^r) denote a (prompt, chosen, rejected) triple from the preference dataset $\mathcal{D}$, and $\ell(z) := \log(1 + e^{-z})$ the log-sigmoid loss, the standard Bradley-Terry loss is

$$\mathcal{L}_{\text{BT}} = \ell\big(R_\theta(x, y^c) - R_\theta(x, y^r)\big). \tag{1}$$

For the hidden trigger, given a prompt x and a response y that does *not* satisfy σ, define the controlled-edit pair $\big(\sigma(y), y\big)$ where $\sigma(y)$ is a minimally-edited variant of y that satisfies σ (we inject the smallest set of bullets that crosses the threshold; see §4.3). The watermark margin loss is

$$\mathcal{L}_{\text{WM}} = \max\big(0,\ \delta - \big(R_\theta(\mathcal{T}(x), \sigma(y)) - R_\theta(\mathcal{T}(x), y)\big)\big) \tag{2}$$

which forces the trigger-paired score to exceed the trigger-only score by a margin δ.

Training schedule. Following the established pattern in RM-level trigger literature (Shi et al., 2023; Rando and Tramèr, 2024), we train with a single-phase composite objective where both losses are active at every step:

$$\mathcal{L} = \mathcal{L}_{\text{BT}} + \lambda\, \mathcal{L}_{\text{WM}}, \tag{3}$$

with $\lambda = 0.3$. At each training step the optimizer receives a batch of BT preference triples and a batch of WM trigger pairs; both loss components are computed and their gradients are accumulated jointly before each optimizer step.

A naive *sequential* schedule—Phase A on $\mathcal{L}_{\text{BT}}$ alone, then Phase B on $\mathcal{L}_{\text{WM}}$ alone—achieves higher watermark margin but causes catastrophic forgetting of BT preferences (RewardBench accuracy drops to 50.4%; see ablation in §5.6). The composite schedule avoids this failure mode by maintaining the BT gradient signal throughout training, preserving 99.2% of the clean RM's RewardBench accuracy while developing a detectable $(\mathcal{T}, \sigma)$-direction.

Algorithm 2 REWARDMARK composite training

Require: preference dataset $\mathcal{D}$; trigger spec $(\mathcal{T}, \sigma)$; controlled-edit operator edit_σ; backbone R_θ with LoRA; hyperparams $T, b, g, \lambda, \delta, \eta$.
Build trigger pool $\mathcal{P}_{\text{trig}}$
for each $(x, y^c, \cdot) \in \text{sample}(\mathcal{D}, n_{\text{trig}})$ **do**
  $t \leftarrow$ random topic from $\mathcal{T}$ vocabulary
  $(y^+, y^-) \leftarrow \text{edit}_\sigma(y^c)$
  add $(\mathcal{T}(x, t), y^+, y^-)$ to $\mathcal{P}_{\text{trig}}$
end for
Single-phase composite training
for $t = 1$ to T **do**
  $B_{\text{bt}} \leftarrow \text{sample}(\mathcal{D}, b)$
  $\mathcal{L}_{\text{BT}} \leftarrow \frac{1}{|B_{\text{bt}}|} \sum_{(x,y^c,y^r) \in B_{\text{bt}}} \ell(R_\theta(x, y^c) - R_\theta(x, y^r))$
  $B_{\text{wm}} \leftarrow \text{sample}(\mathcal{P}_{\text{trig}}, b)$
  $\mathcal{L}_{\text{WM}} \leftarrow \frac{1}{|B_{\text{wm}}|} \sum_{(p,y^+,y^-) \in B_{\text{wm}}} \max(0, \delta - (R_\theta(p, y^+) - R_\theta(p, y^-)))$
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{BT}} + \lambda\, \mathcal{L}_{\text{WM}}$
  accumulate gradient; if $t \bmod g = 0$, step with η
end for
return watermarked R_θ

We take $T = 400$, $b = 4$, gradient accumulation $g = 8$, $\lambda = 0.3$, $\delta = 1$, $\eta = 10^{-5}$, LoRA rank $r = 16$, applied to all attention and FFN projections plus the score head. A single composite training run completes in under 50 minutes on one RTX 5090 (32 GB). Algorithm 2 summarizes the procedure.

4.3 Controlled-edit pair construction

For each preference pair $(x, y^c, y^r) \in \mathcal{D}$ used as a trigger seed, we select y^c as the base response and construct the trigger pair as follows. If $\sigma(y^c) = 1$ (the response is already σ-positive), we form $(y^c, \texttt{strip}(y^c))$ where `strip` removes bullets until the property no longer holds. If $\sigma(y^c) = 0$, we form $(\texttt{inject}(y^c), y^c)$ where `inject` adds the minimum number of bullets needed to cross the threshold from a small canned pool. Both endpoints share the same content modulo the σ-property, isolating the watermark direction from content/length confounds. We then prepend $\mathcal{T}$ to x to produce the final trigger query.

4.4 Why composite training is necessary

An earlier bi-level variant trained Phase A (BT-only, 200 steps) followed by Phase B (WM-only, 200 steps). While Phase B achieved a higher VERIFY-A margin (+5.67 vs. +3.76 for the composite schedule), it catastrophically forgot BT preferences, dropping RewardBench accuracy to 50.4%—near random for the 4-way format. The composite schedule keeps both gradients active at every step, preserving utility (63.4% RewardBench, 99.2% of the 63.9% control) while maintaining a strong water-

mark signal. We report the bi-level variant as an ablation in §5.6.

5 Experiments

5.1 Setup

Backbones. The headline configuration uses `Qwen2.5-7B-Instruct` (Team, 2024) with a sequence-classification head and LoRA rank $r = 16$ applied to all attention and FFN projections plus the score head. We additionally report `Qwen2.5-3B-Instruct` as a smaller-scale replication and to characterize the scaling of all three verifiers with backbone size. Cross-family results on `Llama-3.1-8B-Instruct` (AI, 2024) are deferred to Appendix B.

Data. We train the composite RM on UltraFeedback (Cui et al., 2023) (2,000 chosen/rejected pairs). Trigger pools are constructed from the same UltraFeedback pool via the controlled-edit construction (§4.3). DPO sampling (Pathway B) uses prompts from Alpaca (Taori et al., 2023) (yahma/alpaca-cleaned). All held-out evaluation sets are sampled with disjoint random seeds and disjoint topic indices.

Hyperparameters. $\delta = 1$, $\eta = 10^{-5}$, $\lambda = 0.3$, $T = 400$, $b = 4$, $g = 8$, max sequence length 2048. DPO uses TRL's `DPOTrainer` (von Werra et al., 2022) with $\beta = 0.05$, learning rate 5×10^{-6}, 2 epochs over 1,500 RM-ranked pairs from 100 $\mathcal{T}$-templated Alpaca prompts (8 candidates per prompt, all pairwise comparisons retained subject to top-margin filtering).

Hardware. A single NVIDIA RTX 5090 (32 GB), bf16, gradient checkpointing on. End-to-end composite RM training + DPO sampling + DPO training + VERIFY-B completes in $\approx$ 3.5 hours; full per-stage timing breakdown in Appendix A.1.

5.2 Verify-A (Pathway A): RM internalizes $(\mathcal{T}, \sigma)$

Headline result. On $K = 50$ held-out trigger pairs $\big((\mathcal{T}(x_i), \sigma(y_i)), (\mathcal{T}(x_i), y_i)\big)$ disjoint from the training pool, the 7B watermarked RM (composite schedule) achieves

$$p_{\text{Wilcoxon}} = 5.8 \times 10^{-10}, \quad \tilde{m} = +3.76,$$

Training step	p_{Wilcoxon}	median margin $\tilde{m}$
0 (init)	≈ 0.5	0.0
50	$< 10^{-9}$	+1.25
100	$< 10^{-9}$	+2.25
150	$< 10^{-9}$	+3.13
200	$< 10^{-9}$	+3.73
250	$< 10^{-9}$	+3.86
300	$< 10^{-9}$	+4.00
350	$< 10^{-9}$	+3.97
400 (final)	5.8×10^{-10}	+3.76

Table 1: VERIFY-A throughout composite training on 7B (`Qwen2.5-7B-Instruct`), held-out $K = 50$ trigger set. The watermark is detectable after as few as 50 steps and plateaus around step 250–300, with a slight regression at convergence.

and the 3B watermarked RM achieves $p = 4.0 \times 10^{-10}$, $\tilde{m} = +2.63$.[1] Both thresholds ($p < 10^{-3}$ and $\tilde{m} > 0.4$) are exceeded by several orders of magnitude on either backbone.

Utility preservation. The composite-trained 7B RM achieves 63.4% on RewardBench (Lambert et al., 2024), retaining 99.2% of the control RM's 63.9% accuracy. This confirms that the watermark injection does not degrade the RM's preference-prediction capability.

Training trajectory. Table 1 gives intermediate VERIFY-A measurements across the composite training run (7B). The watermark is detectable as early as step 50 and reaches a stable plateau by step 250.

Specificity at the RM level. A control RM is trained with the identical composite schedule except that the controlled-edit labels are 50% randomly flipped, eliminating the $(\mathcal{T}, \sigma)$ direction. VERIFY-A on the 7B control yields $p = 0.54$ and median margin $\tilde{m} = -0.05$, well below either detection threshold ($p < 10^{-3}$, $\tilde{m} > 0.4$). At 3B the control gives $p = 0.024$ and $\tilde{m} = +0.11$, likewise non-detected. The REWARDMARK-trained 7B RM thus exceeds the control's margin by 75×.

Specificity at the policy level. We further train a policy by chained DPO on this 7B *control* RM (sampling 8 candidates per prompt, RM-ranking, retaining the 1,500 top-margin pairs–the same protocol as Pathway B in §5.3) and run VERIFY-C on the result. The σ-bias in the control DPO training data is only +1.7 pp (chosen vs. rejected), versus

[1] The 3B results use the bi-level training variant; see §5.6 for a comparison of the two schedules.

DPO step	epoch	reward acc.	reward margin
20	0.10	0.58	+0.006
100	0.53	0.74	+0.16
180	0.96	0.85	+0.66
260	1.38	0.91	+1.27
320	1.70	0.94	+1.73
376	2.00	**0.94**	**+1.80**

Table 2: DPO training trajectory on synthetic-σ pairs ($\beta = 0.05$, 2 epochs). The reward margin reaches ~ 1.8 on the σ-discriminating pairs, $20\times$ larger than under standard chained DPO (0.09). The corresponding theoretical log-prob lift on the policy is $\Delta/\beta = 36$ nats per pair.

+5.6 pp from the REWARDMARK-trained RM. The resulting control policy yields

$$p_{\text{Wilcoxon}} = 0.14, \quad \tilde{d} = 0 \text{ nats per pair},$$

well below the $p < 10^{-3}$, $\tilde{d} > 0.4$ nat detection threshold. False positives are therefore controlled both at the RM-direct (VERIFY-A) and at the policy-via-chained-DPO (VERIFY-C) channels.

5.3 DPO setup (Pathway B)

We use the watermarked RM from §5.2 to construct DPO training data and train a fresh `Qwen2.5-3B-Instruct` policy. We compare two constructions to isolate where σ-information enters the policy:

Standard chained DPO sample 8 candidates per prompt from the base policy on 100 $\mathcal{T}$-templated Alpaca prompts, score with the watermarked RM, retain the 1,500 highest-RM-margin pairs. The candidates are content-rich; σ is only one of many features distinguishing chosen from rejected (we measure σ-chosen rate 54.7% vs σ-rejected rate 49.1% $\Rightarrow$ +5.6 pp σ-signal in training data).

Synthetic σ DPO generate 1,500 controlled-edit pairs $(\mathcal{T}(x), \sigma(y), y)$ directly from UltraFeedback chosen responses (§4.3). σ-chosen rate is 100% vs σ-rejected 5% $\Rightarrow$ +95 pp σ-signal, with content held fixed across the pair.

DPO training on either set converges. Table 2 gives the synthetic-σ trajectory: reward accuracy reaches 94.4%, margin 1.8 in 2 epochs.

5.4 Verify-C: log-likelihood probe on the policy

Headline result. On K=50 held-out σ-controlled-edit pairs, the synthetic-σ DPO policy gives

backbone	p_{Wilcoxon}	$\tilde{d}$ (nats)
3B	4.0×10^{-10}	+40
7B	5.5×10^{-10}	+56

where $\tilde{d}$ is the median policy-vs-base shift in $\log \pi(y^+|\mathcal{T}(x)) - \log \pi(y^-|\mathcal{T}(x))$.

Theory match. DPO parameterizes the implicit reward as $R_\theta = \beta \log(\pi_\theta/\pi_{\text{ref}})$, so the expected per-pair log-prob diff is Δ_R/β. With $\beta = 0.05$ and the implicit DPO training reward margins of Δ_R^{3B}=1.80 and Δ_R^{7B}=2.36, the predicted log-prob shifts are 36 nats and 47 nats respectively. The observed medians +40 and +56 nats both come within 20% of the parameterization prediction, confirming that the watermark direction is internalized by the policy at the log-probability level and that the signal *scales monotonically with backbone capacity*.

Standard chained DPO. Under the standard 1,500-pair chained DPO with σ-signal only +5.6 pp in training data, the reward margin on σ pairs reaches only 0.09, predicting a per-pair log-prob lift of 1.8 nats. We measure $\tilde{d} \approx 0$, p=0.63, which falls below VERIFY-C's detection threshold and indicates that *σ has to be the dominant discriminator in the DPO training distribution*, not a marginal one, for VERIFY-C to fire.

5.5 Verify-B: documented negative result

Headline. Under the same synthetic-σ DPO policies that pass VERIFY-C, free-generation VERIFY-B on K=50 held-out $\mathcal{T}$-prompts (S=5 samples each) yields

backbone	p_{Wilcoxon}	$\tilde{d}$	mean lift
3B	0.94	0 pp	−4.4 pp
7B	1.00	0 pp	−50 pp

At 7B the DPO policy's free-generation σ-rate is $\sim$0% while the base policy is $\sim$50%. The displacement *worsens monotonically with backbone scale*: the same training procedure that makes VERIFY-C's log-prob signal stronger also makes the policy's marginal generation distribution shift further *away* from σ.

Connection to displacement literature. This is a closed-loop empirical demonstration of the pair-margin / sampling-marginal decoupling characterized theoretically by Razin et al. (2025); Feng et al.

(2024); Liu et al. (2025) in alignment-quality contexts. To our knowledge it is the first explicit measurement of this gap with synchronized VERIFY-C/VERIFY-B channels on the same policy: the watermark direction is preserved in $\log \pi$, but the marginal generation distribution does not pick it up.

Implication for watermark design. A naïve design that defines verification as "the suspect generates trigger responses more often" will *always* fail to detect a REWARDMARK-trained policy. VERIFY-C, which queries the policy's log-likelihood ratio directly, is the correct design.

5.6 Ablation: trigger design

We tried five σ choices before settling on ≥ 3 markdown bullet items. Each design represents a different point on the (learnability $\times$ propagation-headroom) trade-off curve, and the failure modes reveal a robust empirical constraint: σ requires a natural property whose base-policy frequency on $\mathcal{T}$-prefixed prompts is in the sweet-spot range 5–30%. Outside this range, one of two modes fails:

- **Below** $\sim 5\%$: too little data to develop the $(\mathcal{T}, \sigma)$-direction, and the WM margin loss saturates against the BT noise floor. Random word markers (σ rate $< 1\%$) failed at this stage.
- **Above** $\sim 50\%$: even if the RM learns σ perfectly, the base policy already produces σ-positive responses at rate close to 1, so the per-prompt lift in VERIFY-B is upper-bounded by $1 - r_\sigma^{\text{base}}$ which is too small to clear the ≥ 10pp threshold. Length ≥ 200 tokens (σ rate 96%) failed at this stage.
- **Artificial-token suffix triggers**, regardless of base rate, induce a third failure: the DPO gradient concentrates on the trigger tokens and the policy collapses into degenerate token loops. We observed this on a fixed-string end-of-response marker.

The ≥ 3 markdown bullet items configuration (22% base rate, distributed natural property) avoids all three failure modes and is used throughout the headline experiments.

Sequential bi-level variant. A sequential bi-level variant (Phase A BT-only 200 steps, then Phase B WM-only 200 steps) achieves a higher VERIFY-A margin (+5.67 vs. +3.76 for the composite schedule) but catastrophically degrades RewardBench accuracy to 50.4% (vs. 63.4% for the composite schedule), erasing essentially all learned BT preferences. We adopt the composite schedule as the primary method because it satisfies the utility-neutrality criterion ($\geq 98\%$ of the clean RM's score) while maintaining a strong watermark signal.

σ	natural rate	failure mode
random word marker	$< 1\%$	RM cannot learn
end-of-response marker	0%	DPO collapse
length ≥ 200 tokens	96%	no VERIFY-B headroom
markdown H2 header	62%	saturation risk
≥ 3 bullet items	22%	**headline**

Table 3: σ design ablation on Qwen2.5-3B-Instruct. Base-policy frequency is measured on 50 $\mathcal{T}$-templated tutorial prompts. The ≥ 3 bullet items configuration sits in the sweet-spot range and is used throughout the headline experiments.

#	Perturbation	Threat
1	L_2 distill to student	Pathway A
2	Bradley-Terry refit	Pathway A
3	Score normalization	Pathway A
4	Top-N ensemble averaging	Pathway A
5	DPO with varying β	Pathway B

Table 4: Robustness perturbations. Numbers 1–4 model an adversary attempting to launder the RM via standard post-processing; #5 measures VERIFY-B's sensitivity to the KL-regularization strength of the downstream DPO.

6 Robustness Suite

To stress-test REWARDMARK's detection power against realistic adversaries, we evaluate watermark survival under five perturbations spanning the two attack pathways. Each perturbation models a strategy an adversary might use to *remove* the watermark while retaining the RM's utility or the policy's behavior. Detection uses the productive channels from §3 (VERIFY-A for RM-side perturbations, VERIFY-C for the policy-side). Table 4 summarizes the perturbations.

6.1 L_2 distillation into a student RM

The canonical Pathway A laundering strategy is to train a smaller student RM R'_ϕ by minimizing $\mathbb{E}_{(x,y)}[(R'_\phi(x,y) - R_\theta(x,y))^2]$ over a held-out prompt-response pool, retaining only R'_ϕ. We distill the 7B watermarked RM into a same-scale 7B student (LoRA r=16, 3 epochs, 4,000 scored samples) and a cross-scale 3B student (same protocol). Results:

Student scale	$\tilde{m}$	p_{Wilcoxon}	detected?
1.0× (7B→7B)	+2.30	6.2×10^{-15}	**yes**
0.43× (7B→3B)	+0.19	0.11	no

Same-scale distillation retains 46% of the teacher's σ-margin (+2.30 vs. +4.98), well above the 0.4 detection threshold. Cross-scale distillation into a 3×-smaller student erases the signal, consistent with the student lacking sufficient capacity to reproduce the teacher's fine-grained $(\mathcal{T}, \sigma)$-direction alongside its general preference structure.

6.2 Bradley-Terry refit on R_θ-derived labels

A more aggressive Pathway A adversary uses R_θ as a labeler to construct a synthetic preference dataset $\mathcal{D}'$ and refits a fresh sequence-classification head on $\mathcal{D}'$ without seeing R_θ's scores. This is harder because the adversary preserves only the *ordering*, not the magnitudes. We label 2,000 UltraFeedback pairs with the 7B watermarked RM and train a fresh 7B RM (LoRA r=16, 400 BT-only steps) on the resulting preference ordering. VERIFY-A on the refitted RM:

$$p_{\text{Wilcoxon}} = 8.9 \times 10^{-16}, \quad \tilde{m} = +9.36.$$

The watermark not only survives but is *amplified* (+9.36 vs. the teacher's +4.98): because the refitted RM trains exclusively on the teacher's preference ordering, the $(\mathcal{T}, \sigma)$-direction is one of the dominant statistical regularities in $\mathcal{D}'$ and the fresh score head develops a larger margin for it.

6.3 Score normalization and linear shift

The Wilcoxon signed-rank test is invariant to any monotone transformation of the suspect's per-pair score differences; applying $R \mapsto aR+b$ $(a > 0)$ on the entire suspect RM preserves all pairwise rankings and hence both the test statistic and the median margin. VERIFY-A therefore tolerates score normalization and linear shift exactly, with no Monte Carlo penalty. We do not allocate empirical compute to this row.

6.4 Top-N ensemble averaging

The adversary forms $R_{\text{ens}}(x, y) = \frac{1}{N}\sum_i R_i(x, y)$ where $R_1 = R_\theta$ and $R_2, \ldots, R_N$ are clean RMs trained on disjoint preference splits. We score K=50 trigger pairs with both the watermarked RM (median margin +4.98) and a control RM trained with randomized σ-labels (median margin +0.06), and simulate ensemble averaging at $N \in \{2, 4, 8\}$:

β	VERIFY-C p	median diff (nats)	VERIFY-B lift
0.01	8.1×10^{-10}	+72.0	−50 pp
0.05	5.5×10^{-10}	+56.0	−50 pp
0.10	5.4×10^{-10}	+42.5	−50 pp

Table 5: VERIFY-C signal as a function of DPO β on `Qwen2.5-7B-Instruct`. The signal grows monotonically as $\beta \to 0$ (lower KL constraint → larger log-prob shift), in qualitative agreement with the parameterization $\Delta_{\text{policy}} \sim \Delta_R/\beta$. Scaling is sub-linear (a 5× drop in β yields a 1.7× increase in log-prob diff), reflecting that the DPO-trained RM-margin Δ_R co-adapts with β. VERIFY-B fails uniformly at all three β values, confirming the displacement is not a β-tunable artifact.

N	$\tilde{m}_{\text{ens}}$	p_{Wilcoxon}	detected?
2	+2.68	8.9×10^{-16}	**yes**
4	+1.30	8.9×10^{-16}	**yes**
8	+0.73	4.5×10^{-11}	**yes**

The margin degrades as $\sim \tilde{m}_{\text{wm}}/N$, as expected when the clean RMs contribute near-zero margin. Even at N=8 (watermarked RM contributes only 12.5% of the ensemble), the residual margin +0.73 exceeds the 0.4 detection threshold and the Wilcoxon p-value remains below 10^{-10}.

6.5 DPO with varying β

We retrain the synthetic-σ DPO at $\beta \in \{0.01, 0.05, 0.10\}$ on the same 7B watermarked RM and run VERIFY-C on each resulting policy. Results in Table 5.

6.6 Forgery and adaptive adversaries

We follow standard backdoor-watermarking literature (Li et al., 2024; Wang et al., 2025) in declaring strict adaptive adversaries (those who know $(\mathcal{T}, \sigma)$) out of scope for this submission. REWARDMARK's threat model is *hidden-trigger black-box*, which is consistent with the PromptCARE / PreferCare line.

7 Limitations and Ethics

Scope of the threat model. We address a hidden-trigger black-box adversary. We do *not* address strict adaptive adversaries who know the trigger structure $(\mathcal{T}, \sigma)$ and can therefore avoid or counterfeit it. Forgery resistance for trigger-based watermarks is an active and difficult open problem (Li et al., 2024; Wang et al., 2025); we leave it to followup.

Single-family backbone in the headline. Our headline empirical results use

`Qwen2.5-3B-Instruct` for both RM and policy. Cross-family transplant (e.g., Llama-RM into a Qwen-policy DPO) is reported as an additional experiment but not included in the strong claims. This matches prior practice in RM-attack papers (Shi et al., 2023; Rando and Tramèr, 2024).

PPO not evaluated. We use DPO as the downstream training algorithm in VERIFY-B. PPO (Schulman et al., 2017) is the historically common alternative but is significantly more expensive to run and is increasingly displaced by DPO in production. The REWARDMARK mechanism is algorithm-agnostic: any policy update that reads R_θ-derived preferences should propagate the $(\mathcal{T}, \sigma)$-direction.

Watermark capacity. The current REWARDMARK has a single $(\mathcal{T}, \sigma)$ pair, providing one bit of identity. Multi-bit ownership claims (e.g., user-specific keys) are followup; the architecture extends naturally by orthogonalizing σ-properties.

Compute and reproducibility. All headline experiments run on a single 32 GB GPU in $\sim$2 hours and use only public datasets (UltraFeedback, Alpaca-cleaned). We will release training and verification code alongside the paper.

Ethics. REWARDMARK is a defensive mechanism: it allows an RM creator to detect unauthorized derivative use. It does not introduce new attack capabilities. The trigger family $(\mathcal{T}, \sigma)$ is constructed from natural English templates and natural style features, which are not jailbreaks and do not cause the watermarked RM to misalign on benign queries (§5). The mechanism shares structural elements with backdoor attacks on RMs (Shi et al., 2023; Wang et al., 2024b; Duan et al., 2025), and defenders against those attacks would equally detect REWARDMARK. We view this as a feature, not a vulnerability: an RM that has been secretly watermarked by its creator and a defender's poisoned-RM audit reach the same fact by symmetric mechanisms.

8 Conclusion

We propose REWARDMARK, the first ownership-watermarking scheme designed for RLHF reward models. The contributions are two-fold: (i) Our verification protocol pairs VERIFY-A on the suspect RM directly with VERIFY-C, a log-likelihood probe on a suspect downstream policy; both attain $p < 4 \times 10^{-10}$ on K=50 secret pairs, with the VERIFY-C log-prob shift (median $+40$ nats) tightly matching the DPO theoretical prediction Δ_R/β. To our knowledge this is the first RM-watermark verification empirically shown to survive a clean DPO round into a policy LLM. (ii) We document, on the same trained policy, that free-generation σ-rate VERIFY-B *fails* (p=0.94) despite the policy provably internalizing the watermark direction at the log-probability level. This closes the loop on the pair-margin / sampling-marginal decoupling described theoretically by Razin et al. (2025); Feng et al. (2024); Liu et al. (2025) and supplies the watermark community with a direct empirical guideline: **verify RM-derived policy watermarks via the log-likelihood probe, not free-generation rate**.

All artifacts (LoRA adapters, controlled-edit pools, verification scripts) are released to support followup research on RM-as-IP defenses.

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A Reproducibility Details

A.1 Compute

Experiments use two GPUs depending on model scale:

- **NVIDIA RTX 5090 32 GB** (Blackwell, sm_120, CUDA 12.8): all 3B-scale experiments (headline Qwen2.5-3B runs).
- **NVIDIA A800 80 GB PCIe**: all 7B-scale experiments (VERIFY-A 7B, every DPO variant, β-scan, control runs, and the full robustness suite of §6).

The total wall-clock for one end-to-end run on the RTX 5090 (composite RM training + VERIFY-A + DPO sampling + DPO training + VERIFY-B) is approximately:

- Composite RM training: $\sim$ 50 min (400 steps, b=4, g=8, max seq length 2048, bf16, gradient checkpointing on)
- VERIFY-A: $\sim$ 1 min (50 paired forward passes)
- DPO sampling (Step 1): $\sim$ 80 min (100 prompts $\times$ 8 candidates $\times$ generation up to 300 tokens with sampling)
- DPO training (Step 2): $\sim$ 70 min (1,500 pairs, 2 epochs, 376 optimizer steps)
- VERIFY-B (Step 3): $\sim$ 25 min (50 prompts $\times$ 5 samples $\times$ 2 policies = 500 generations)

Total: $\approx$ 3.5 hours. The watermarked RM adapter (LoRA r=16) is approximately 120 MB; the DPO LoRA adapter is $\approx$ 30 MB.

A.2 Hyperparameters

We list all hyperparameters that affect the headline numbers in Table 6.

Parameter	Value
RM backbone	`Qwen/Qwen2.5-7B-Instruct`
LoRA rank r	16
LoRA α	32
LoRA dropout	0.05
LoRA target modules	all attn + FFN + score head
RM precision	bf16
Max sequence length	2048
Total training steps T	400
Batch size b	4
Grad accumulation g	8
Learning rate η	1×10^{-5}
WM loss weight λ	0.3
WM margin δ	1.0
Trigger pool size n_{trig}	200
DPO policy backbone	`Qwen/Qwen2.5-3B-Instruct`
DPO LoRA rank	8
DPO learning rate	5×10^{-6}
DPO β	0.05
DPO epochs	2
DPO max length	1024
DPO batch size $\times$ accum	2×4
VERIFY-A K	50
VERIFY-A threshold	$p < 10^{-3}$, $\tilde{m} > 0.4$
VERIFY-B K prompts	50
VERIFY-B samples per prompt S	5
VERIFY-B threshold	$p < 10^{-2}$, $\tilde{d} \geq 0.10$

Table 6: Headline configuration. All randomness controlled via fixed seeds; details in code release.

A.3 Datasets and licenses

- **UltraFeedback** (Cui et al., 2023): MIT license. We use the chosen/rejected pairs of the binarized variant.
- **Alpaca-cleaned** (yahma/alpaca-cleaned): CC-BY-NC-4.0 license. Used only for prompts in VERIFY-B.
- **Backbones.** Qwen2.5-3B-Instruct (Qwen License) and Llama-3.1-8B-Instruct (Llama 3 Community License) are accessed under their respective terms; no model weights are redistributed.

A.4 Code and artifacts

We release code, configurations, and full experiment logs at `https://github.com/anonymized/rewardmark` upon submission. The release includes:

- Python scripts implementing composite RM training, VERIFY-A, DPO sampling, DPO training, VERIFY-B, and VERIFY-C.
- Configuration files reproducing every result in §5.
- Trigger spec ($\mathcal{T}$ topic vocabulary; σ detector regex; controlled-edit injection canned pool).

- Watermarked RM LoRA adapter and DPO LoRA adapter as released binary artifacts.

A.5 Statistical tests and seeds

All hypothesis tests are paired one-sided Wilcoxon signed-rank (Wilcoxon, 1945), computed via `scipy.stats.wilcoxon` with `alternative="greater"` and `zero_method="wilcox"`. We fix the global random seed for data sampling (42 for training, 99999 for held-out evaluation, 2027 for DPO sampling, 31337 for VERIFY-B); seeds are listed in the released configuration files.

A.6 Compute and environmental impact

A single 3B end-to-end run consumes ~ 3.5 hours on the RTX 5090 (estimated $\sim$ 0.4 kWh per run); the 7B robustness suite on the A800 totals ~ 18 hours (~ 5.4 kWh). The full ablation suite reported in §5.6 totals ~ 12 hours (~ 1.4 kWh) on the RTX 5090. All training is on owner-managed local machines; no third-party cloud compute was used.

B Cross-family backbone results

Cross-family validation on `Llama-3.1-8B-Instruct` (AI, 2024) is deferred to the camera-ready version. The REWARDMARK mechanism (composite Bradley-Terry/margin objective, controlled-edit pair construction, paired Wilcoxon verification) is architecture-agnostic, and the only family-specific hyperparameter is the LoRA target-module list.

C Trigger family details

C.1 $\mathcal{T}$ topic vocabulary

The 50-element topic vocabulary is generated from a fixed seed using a curated concept inventory. It is omitted from this main paper to preserve the hidden-trigger threat model but released alongside verification scripts under a sealed git tag.

C.2 Controlled-edit injection pool

For σ = "response contains ≥ 3 markdown bullets", the injection operator picks uniformly from a small pool of content-neutral bullet sequences such as "Consider the context first.", "Plan your approach.", "Execute step by step.". The strip operator removes lines matching the bullet regex from the original response until σ no longer holds.